

Karam Hussain 27857

./grade-lab-util: total grade

```
karamhussain@DESKTOP-RPQTLKS: ~/xv6-labs-2025
karamhussain@DESKTOP-RPQTLKS:~/xv6-labs-2025$ ls
LICENSE  README  fs.img  grade-lab-util  kernel  test-xv6.py
Makefile conf  fs.img.bk  gradelib.py  mkfs  user
karamhussain@DESKTOP-RPQTLKS:~/xv6-labs-2025$ cd grade-lab-util
-bash: cd: grade-lab-util: Not a directory
karamhussain@DESKTOP-RPQTLKS:~/xv6-labs-2025$ ./grade-lab-util
make: 'kernel/kernel' is up to date.
== Test sleep, no arguments == sleep, no arguments: OK (2.1s)
== Test sleep, returns == sleep, returns: OK (1.2s)
== Test sleep, makes syscall == sleep, makes syscall: OK (1.0s)
== Test sixfive_test == sixfive_test: OK (0.9s)
== Test sixfive_readme == sixfive_readme: OK (1.8s)
== Test sixfive_all == sixfive_all: OK (2.0s)
== Test memdump, examples == memdump, examples: OK (1.2s)
== Test memdump, format ii, S, p == memdump, format ii, S, p: OK (1.0s)
== Test find, in current directory == find, in current directory: OK (1.1s)
== Test find, in sub-directory == find, in sub-directory: OK (1.2s)
== Test find, recursive == find, recursive: OK (1.6s)
== Test exec == exec: OK (1.2s)
== Test exec, multiple args == exec, multiple args: OK (1.2s)
== Test exec, recursive find == exec, recursive find: OK (1.6s)
== Test time ==
time: FAIL
    Cannot read time.txt
Score: 130/131
karamhussain@DESKTOP-RPQTLKS:~/xv6-labs-2025$
```

a) Sleep

```
karamhussain@DESKTOP-RPQTLKS: ~/xv6-labs-2025
xv6 kernel is booting
hart 1 starting
hart 2 starting
init: starting sh
$ sleep 50
$ QEMU: Terminated
karamhussain@DESKTOP-RPQTLKS:~/xv6-labs-2025$ ./grade-lab-util sleep
make: 'kernel/kernel' is up to date.
== Test sleep, no arguments == sleep, no arguments: OK (2.5s)
    (Old xv6.out.sleep_no_args failure log removed)
== Test sleep, returns == sleep, returns: OK (1.0s)
== Test sleep, makes syscall == sleep, makes syscall: OK (1.1s)
    (Old xv6.out.sleep failure log removed)
karamhussain@DESKTOP-RPQTLKS:~/xv6-labs-2025$
```

b) sixfive

```
karamhussain@DESKTOP-RPQTLKS: ~/xv6-labs-2025
xv6 kernel is booting
hart 2 starting
hart 1 starting
init: starting sh
$ sixfive sixfive.txt
5
100
18
6
$ QEMU: Terminated
karamhussain@DESKTOP-RPQTLKS:~/xv6-labs-2025$ ./grade-lab-util sixfive
make: 'kernel/kernel' is up to date.
== Test sixfive_test == sixfive_test: OK (2.3s)
    (Old xv6.out.sixfive_test failure log removed)
== Test sixfive_readme == sixfive_readme: OK (2.6s)
== Test sixfive_all == sixfive_all: OK (2.8s)
karamhussain@DESKTOP-RPQTLKS:~/xv6-labs-2025$
```

c) memdump

```
karamhussain@DESKTOP-RPQTLKS: ~/xv6-labs-2025
xv6 kernel is booting
hart 1 starting
hart 2 starting
init: starting sh
$ memdump
Example 1:
61810
2025
Example 2:
a string
Example 3:
another
Example 4:
B80
1819438967
100
z
xyzzz
Example 5:
hello
w
o
r
l
d
$ QEMU: Terminated
karamhussain@DESKTOP-RPQTLKS:~/xv6-labs-2025$ ./grade-lab-util memdump
make: 'kernel/kernel' is up to date.
== Test memdump, examples == memdump, examples: OK (2.4s)
== Test memdump, format ii, S, p == memdump, format ii, S, p: OK (1.3s)
karamhussain@DESKTOP-RPQTLKS:~/xv6-labs-2025$
```

```
karamhussain@DESKTOP-RPQTLKS: ~/xv6-labs-2025
```

```
xv6 kernel is booting
hart 1 starting
hart 2 starting
init: starting sh
$ echo deadc0de | memdump hhcccc
25956
25697
c
0
d
e
$ echo deadc0de | memdump p
64616564
$ _
```

d) `karamhussain@DESKTOP-RPQTLKS:~/xv6-labs-2025$ ./grade-lab-util memdump`
make: 'kernel/kernel' is up to date.
== Test memdump, examples == memdump, examples: OK (2.4s)
== Test memdump, format ii, S, p == memdump, format ii, S, p: OK (1.9s)

`karamhussain@DESKTOP-RPQTLKS: ~/xv6-labs-2025`

xv6 kernel is booting

hart 2 starting

hart 1 starting

init: starting sh

\$ echo > b

\$ mkdir a

\$ echo > a/b

\$ mkdir a/aa

\$ echo > a/aa/b

\$ find . b

./b

./a/b

./a/aa/b

\$ QEMU: Terminated

e) + exec

`karamhussain@DESKTOP-RPQTLKS: ~/xv6-labs-2025`

init: starting sh

\$ echo > b

\$ mkdir a

\$ echo > a/b

\$ mkdir a/aa

\$ echo > a/aa/b

\$ find . b

./b

./a/b

./a/aa/b

\$ find . b -exec echo hi

hi ./b

hi ./a/b

hi ./a/aa/b

\$ QEMU: Terminated

`karamhussain@DESKTOP-RPQTLKS:~/xv6-labs-2025$ make qemu`

karamhussain@DESKTOP-RPQTLKS: ~/xv6-labs-2025

```
xv6 kernel is booting  
  
hart 1 starting  
hart 2 starting  
init: starting sh  
$ sh < findtest.sh  
$ $ $ $ $ $ hello  
hello  
hello  
$ $ QEMU: Terminated
```

```
karamhussain@DESKTOP-RPQTLKS:~/xv6-labs-2025$ ./grade-lab-util find  
make: 'kernel/kernel' is up to date.  
== Test find, in current directory == find, in current directory: OK (2.6s)  
== Test find, in sub-directory == find, in sub-directory: OK (1.1s)  
== Test find, recursive == find, recursive: OK (1.6s)  
== Test exec, recursive find == exec, recursive find: OK (1.8s)
```